**Ex No: 3b**

## IMPLEMENTION OF A FUZZY INFERENCE SYSTEM

**Aim*:***

To implement Fuzzy Inference System.

**Procedure:**

**Program:**

Example: Simple Fuzzy Inference System for Temperature Control

```
import numpy as np
import skfuzzy as fuzz
import matplotlib.pyplot as plt

# Step 1: Define the fuzzy variables (input: temperature, output: fan speed)
temperature = np.arange(0, 41, 1)  # Temperature from 0 to 40 degrees Celsius
fan_speed = np.arange(0, 11, 1)  # Fan speed from 0 to 10

# Step 2: Define fuzzy membership functions
# Temperature membership functions: Low, Medium, High
temp_low = fuzz.trimf(temperature, [0, 0, 20])
temp_medium = fuzz.trimf(temperature, [10, 20, 30])
```

```
temp_high = fuzz.trimf(temperature, [20, 30, 40])

# Fan speed membership functions: Low, Medium, High
fan_low = fuzz.trimf(fan_speed, [0, 0, 5])
fan_medium = fuzz.trimf(fan_speed, [2, 5, 8])
fan_high = fuzz.trimf(fan_speed, [5, 10, 10])

# Step 3: Visualize the membership functions
plt.figure(figsize=(10, 6))

plt.subplot(2, 1, 1)
plt.plot(temperature, temp_low, label='Low')
plt.plot(temperature, temp_medium, label='Medium')
plt.plot(temperature, temp_high, label='High')
plt.title("Temperature Membership Functions")
plt.xlabel("Temperature (°C)")
plt.ylabel("Membership Degree")
plt.legend()

plt.subplot(2, 1, 2)
plt.plot(fan_speed, fan_low, label='Low')
plt.plot(fan_speed, fan_medium, label='Medium')
plt.plot(fan_speed, fan_high, label='High')
plt.title("Fan Speed Membership Functions")
plt.xlabel("Fan Speed")
plt.ylabel("Membership Degree")
plt.legend()

plt.tight_layout()
plt.show()
```

```
# Step 4: Fuzzification - Define a crisp temperature input
temperature_input = 28  # Example temperature

# Fuzzify the input
temp_low_level = fuzz.interp_membership(temperature, temp_low, temperature_input)
temp_medium_level = fuzz.interp_membership(temperature, temp_medium, temperature_input)
temp_high_level = fuzz.interp_membership(temperature, temp_high, temperature_input)

# Step 5: Apply fuzzy rules
# Rule 1: If temperature is Low, fan speed is Low
# Rule 2: If temperature is Medium, fan speed is Medium
# Rule 3: If temperature is High, fan speed is High

fan_low_level = temp_low_level
fan_medium_level = temp_medium_level
fan_high_level = temp_high_level

# Step 6: Defuzzification - Calculate the crisp output (fan speed)
fan_output = fuzz.defuzz(fan_speed, fan_low * fan_low_level + fan_medium * fan_medium_level + fan_high * fan_high_level, 'centroid')

# Step 7: Print the output
print(f"Temperature: {temperature_input}°C")
print(f"Fuzzified fan speed: {fan_output:.2f}")
```

**Output**

```
Temperature: 28°C
Fuzzified fan speed: 6.00
```

Output

```
Temperature: 28°C
Fuzzified fan speed: 6.00
```